Arwin Karir

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Education

University of Toronto September 2025 – May 2029
BASc in Computer Engineering (PEY Co-op) *Toronto, ON*

- **Relevant Coursework:** Linear Algebra, Programming Fundamentals (C++, Data Structures), Computer Fundamentals (C, Algorithms), Digital Systems

Experience

Software Engineer May 2026 – Present
SellStatic *Toronto, ON*

- Designed a fault-tolerant AI-video completion pipeline (AWS SNS, SQS with dead-letter queue, Lambda, SES; Azure Blob Storage) that delivers generated videos to users who leave the app mid-render
- Shipped revision history for AI-generated ads with backend validation, durable storage, blob-cleanup retries, and stale-credit recovery, fixing failure cases in a deployed generative-AI product
- Built a brand-onboarding flow (React, C# ASP.NET Core) that imports a company's logo and extracts its color palette to configure AI ad generation, covered by 10+ tests
- Built a GA4 analytics layer with a standardized event taxonomy tracking clicks, feature timing, credit spend, and subscriptions across 27 product surfaces to measure user experience

Artificial Intelligence Intern June 2023 – August 2023
Amazon Work Experience Program *Remote*

- Built a supervised machine-learning pipeline in Python (pandas, NumPy, scikit-learn) to classify 7 forest cover types from 54 environmental features across 464,811 records, min-max scaling features whose ranges spanned 3 to 3,000
- Fixed a 104× class imbalance (2,179 to 226,793 samples per class) by oversampling all 7 classes with SMOTE
- Tuned a decision tree with Optuna (max depth 40, entropy) to 93.4% accuracy, beating a neural network (70.6%) and logistic regression (61.7%); used Matplotlib feature importance to find the top predictor
- Benchmarked training time across all 3 models (6.6 s to 285.7 s) on AWS (SageMaker, S3, EC2) and presented accuracy-versus-cost trade-offs to AWS engineers as part of a 4-person team

Projects

CUDA Optimization of Fused Causal Softmax | *CUDA, C++, PyTorch* July – August 2026

- Wrote a custom PyTorch GPU extension in CUDA C++ that fuses scaling, causal masking, numerically stable exponentiation, reduction, and normalization into a single block-parallel FP32 kernel
- Outperformed PyTorch eager softmax by 1.40–3.94× on an NVIDIA T4 (sequence lengths 128–2048), beat torch.compile-generated kernels on 6 of 7 shapes, and sped up transformer attention by 1.54–2.31× end to end
- Validated 88 numerical and edge-shape test cases (max absolute error 3.58×10^{-7}) and profiled kernels with Nsight Compute, PyTorch Profiler, and CUDA events in commit-tagged benchmarking workflows

LumiSense AI | *PyTorch, ONNX, OpenCV, Raspberry Pi* January 2026 – Present

- Building a computer-vision camera system on Raspberry Pi for the UofT Sustainability Office that estimates floor-level illuminance, transmits and stores lighting data, and estimates carbon emissions from lighting
- Designed a multitask encoder–decoder model (ResNet-style backbone, U-Net-style upsampling) with 3 heads for segmentation, lux regression, and uncertainty, producing dense lux maps and point-wise lux queries
- Achieved 3.97% average error (4.13% max) against benchmark lux readings at 13 spots under and between fixtures
- Added luminance-based frame-quality gating that rejects unusable camera frames before inference, plus on-device debug logging and snapshot capture for deployment reliability

Languages & Technologies

Languages: Python, C/C++, CUDA, C#, TypeScript, JavaScript, SQL
ML & Computer Vision: PyTorch, ONNX Runtime, OpenCV, scikit-learn, pandas, NumPy, Optuna, Matplotlib
Performance & Tools: Nsight Compute, PyTorch Profiler, torch.compile, Git, Linux, Docker, Raspberry Pi
Cloud & Data: AWS (SageMaker, S3, EC2, Lambda, SQS, SNS), Azure, PostgreSQL, MongoDB
Web: React, Next.js, ASP.NET Core, REST APIs, Playwright